The Midnight Library Mystery

Whodunnit? · Foundations of Algebra

Name: _____

It was Family Game Night at the Westbrook Public Library when the lights flickered. When they came back on, the glass case in the lobby was empty: **The Map of Everything**, a 200-year-old atlas, was gone!

Six people were in the building, each seen in a different room, and there are six sneaky ways to smuggle out a giant book. You are the lead detective. Find out **who** took it, **where** they were, and **how** they got it away.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects

- ☐ Mr. Patel — Head Librarian
- ☐ Ms. Greene — Book Club Leader
- ☐ Officer Ruiz — Night Guard
- ☐ Jada Brooks — Teen Volunteer
- ☐ Professor Hale — Visiting Author
- ☐ Mr. Fox — Game Night Host

The Places

- ☐ Front Desk
- ☐ Reading Room
- ☐ Rare Books Vault
- ☐ Children’s Corner
- ☐ Map Room
- ☐ Computer Lab

The Methods

- ☐ Hidden in a book cart
- ☐ Tucked in a backpack
- ☐ Slipped into a board-game box
- ☐ Rolled up in a poster tube
- ☐ Swapped with a fake cover
- ☐ Lowered out a window in a basket

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Midnight Library Mystery · Show every step in the matching box. Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Midnight Library Mystery

Evaluate:

$$18 - 2(3 + 4)$$

A. 112	Professor Hale was signing books at the Front Desk the whole time.
B. 16	The poster tubes were checked. Only posters inside.
C. 4	Mr. Patel was checking out books at the Front Desk the whole time. Twelve families saw him there.
D. 23	Mr. Patel took a long break in the Map Room.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Midnight Library Mystery

Evaluate:

$$3^2 + 4 \cdot 5 - 6$$

A. 23	The book cart was searched at the door. It held only picture books.
B. 20	Professor Hale never left the Reading Room.
C. 59	Jada Brooks was seen hurrying out of the Map Room.
D. 5	Officer Ruiz was patrolling the Computer Lab.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Midnight Library Mystery

Evaluate when x = 5:

$$4x - 7$$

A. 38	A guard checked every poster tube at the door.
B. 13	Ms. Greene was leading a book club in the Reading Room. Her group never let her leave.
C. -8	Ms. Greene was alone in the Map Room for ten minutes.
D. 27	The Map Room was locked all evening.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Midnight Library Mystery

Evaluate when a = −3 and b = 4:

$$a^2 - 2b$$

A. -17	Mr. Fox carried a large backpack out to his car.
B. -14	Professor Hale spent the night in the Children’s Corner.
C. 17	Someone saw a book cart rolling toward the loading dock.
D. 1	Every backpack was checked at the exit. No atlas would even fit.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Midnight Library Mystery

Simplify:

$$5x + 3 - 2x + 8$$

A. $14x$	Officer Ruiz left the vault to check the Map Room.
B. $7x + 11$	Professor Hale was in the Rare Books Vault with the guard.
C. $3x + 5$	The window basket was found sitting on the lawn.
D. $3x + 11$	Officer Ruiz was locked inside the Rare Books Vault doing inventory. The vault log shows the door never opened.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Midnight Library Mystery

Simplify:

$$4(2x - 3)$$

A. $8x - 3$	Jada Brooks had a poster tube under her arm.
B. $6x - 7$	Professor Hale was helping in the Computer Lab.
C. $8x - 12$	The board-game boxes were all opened for Game Night. Nothing but game pieces inside.
D. $8x + 12$	A board-game box felt much too heavy.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Midnight Library Mystery

Simplify:

$$3(x + 4) - 5x$$

A. $8x + 12$	Mr. Patel walked out with a very large backpack.
B. $-2x + 4$	Professor Hale was reading in the Children’s Corner.
C. $2x + 12$	The fake cover had been swapped with the real one.
D. $-2x + 12$	Jada Brooks was reading to toddlers in the Children’s Corner. A dozen parents took videos.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Midnight Library Mystery

Write an expression:

5 less than 3 times a number x

A. $5 - 3x$	Mr. Fox was spotted near the Map Room door.
B. $3(x - 5)$	Ms. Greene asked to borrow a poster tube.
C. $3x - 5$	The fake cover in the case is a decoy the library always uses. The real atlas was never swapped.
D. $5x - 3$	Professor Hale never touched a poster tube.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Midnight Library Mystery

Evaluate:

$$|-8| + (-3)(4)$$

A. -20	Officer Ruiz left the vault for an hour.
B. 20	The poster tube was empty when security looked.
C. 9	Jada Brooks left the Children’s Corner early.
D. -4	Mr. Fox was running a trivia contest from the Computer Lab. The livestream never shows him leaving.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

The Midnight Library Mystery

Simplify:

$$-2(3x - 5) + 7x$$

A. $x - 10$	A basket was lowered from the Reading Room window.
B. $x + 10$	All the windows are painted shut for the winter. Nothing went out in a basket.
C. $13x + 10$	Professor Hale was at the Front Desk with Mr. Patel.
D. $x + 5$	Mr. Fox left the livestream for twenty minutes.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.